

Examiner reports

Q1.

Here the similarity of the answers resulted in many students making the wrong choice. The correct answer was most popular, but was only chosen by 35%. The second favourite was option 1, suggesting that many calculated the gradient of the radius from the origin to the centre rather than the tangent at the origin.

Q2.

In part (a), about two thirds of all students were able to draw the required two circles correctly. There were some poor 'hand-drawn' circles which required some liberal interpretation from examiners. It would be preferable if students use a pair of compasses when required to draw a circle.

In part (b) fewer than expected recognised or stated that the y -coordinate of the centre was $y = 5$, and fewer still were able to form an equation to find the possible x -coordinates of the centre: $\pm\sqrt{11}$. Some worked with the general equation of circles which proved a successful strategy. Any correct form of the final two circles was accepted.

Q3.

In part (a) students had a choice of techniques to use and the first mark was awarded for using an appropriate technique to solve the problem. This was a "show that" question, which was an indication that a rigorous argument needed to be given. Those who compared gradients often failed to complete their argument with a statement such as " $m_1 \times m_2 = -1$ means that the lines are perpendicular, so ABC is a right angle."

Part (b)(i) was intended to help students to answer part (b)(ii). While most students seemed to know the correct theorem, it was usually poorly stated.

Part (b)(ii) was generally done well, with students making good progress. The main errors seen here were numerical slips or failing to give a justification for their answer. Many students showed clear working, gave an argument to show the distance from the centre to D was greater than the radius and concluded that D lay outside the circle.

Q4.

This question gave students complete choice as to how to go about solving the problem. Many good attempts were seen, with the full range of marks being scored. Different correct approaches were used and given credit. For example, some students modelled using the length and width of the rectangle, others used a radius and angle to form an expression for area using trigonometry.

Marks were often lost through students not fully justifying their answer. Some students failed to identify their variables or were inconsistent in their use. Others did not explain why they had differentiated or how they knew they had found a maximum.

Q5.

The vast majority of students answered this multiple choice question correctly, giving the

answer 9π .

Q8.

- (a) Many students began by trying to complete the square for both the x terms and the y terms. These students, in an attempt to put $x = 0$, often simply wrote $(y - 2)^2 = 12$ and scored no marks. A surprisingly small number of students substituted $x = 0$ directly into the original equation; those who followed this method sometimes stopped at $y^2 - 4y = 12$, apparently being unable to solve the quadratic equation; others concluded that “ $y = 12$ or $y - 4 = 12$ ”, which is a bad error and not really expected at this level of mathematics. Those trying to solve $(y - 2)^2 = 16$ tended to miss the negative square root of 16.

Students who found that the radius was 5 sometimes used a diagram with a right angled triangle, two of the sides being 3 and 5. This enabled them to find the y -intercepts using the y -coordinate of C , by writing $y = 2 \pm 4$.

- (b) In their working for the previous part, many had already established the equation of the circle in the form $(x + 3)^2 + (y - 2)^2 = 25$ and could write down the value of the radius. However, the correct method was in doubt when 25^2 , or $\sqrt{25}$, instead of 25, appeared on the right hand side of the student's equation, even though in each of these cases the radius was then stated as being 5.

A significant number of students, despite the coordinates of the centre being given in the question, wrote down an incorrect circle equation such as $(x - 3)^2 + (y + 2)^2 = 25$ and once again concluded that the radius was equal to 5. Students cannot expect to score full marks for any answer that follows incorrect working.

- (c) (i) Although most students recognised that they should use the distance formula, many added 25 and 9 to make 36 not 34. There were also many examples of poor notation, such as $5^2 + 3^2 = \sqrt{34}$. There were many arithmetic slips due to the negative coordinate of C . Those who used an incorrect formula for the length of CP scored no marks at all.
- (ii) Some realised that $(2, 2)$ was one of the points of contact and then found PQ very easily. Failing to recognise the position of the right angle often led to many students writing $\sqrt{34 + 25}$ for the length of PQ . Many, who perhaps believed that they were using Pythagoras correctly, wrote $\sqrt{34} - \sqrt{25} = \sqrt{9}$ and this scored no marks, even though students obtained the correct value for the length of PQ . The mention of “tangent” often triggered attempts to find equations rather than using the geometrical properties of the circle.

Q9.

- (a) Most candidates began by trying to complete the square for both the x terms and the y terms and many were successful. However, several gave the left hand side of the equation for the circle as $(x - 5)^2 + (y - 7)^2$ or $(x + 5)^2 + (y - 7)^2$ and some omitted the plus sign between the squared terms or had a minus sign instead. It was not uncommon to see a negative value on the right hand side of the circle equation and 25 was often seen instead of 49.

- (b) (i) This was usually correct, or followed from the candidate's circle equation.
- (ii) Provided the candidate's value of k in their circle equation was positive, candidates were given credit for stating $\sqrt{k}$ as the value for the radius.
- (c) (i) Candidates usually drew a circle where the centre was approximately in the correct quadrant. Less attention was paid to where the circle cut or touched the axes. It was necessary to show the circle touching the x -axis at (5,0) and cutting the negative y -axis twice. A few candidates neglected to draw any axes at all and scored no marks.
- (ii) Those who had found the correct centre and radius usually had the correct coordinates. Some simply wrote pairs of coordinates on their circle but often failed to identify the coordinates of the point that was furthest away from the x -axis.
- (d) The majority of candidates knew that the transformation was a "translation" although a few included a further transformation such as a "stretch" and some used incorrect words such as "shift". However, very few stated the correct column vector and $\begin{bmatrix} -6 \\ 7 \end{bmatrix}$ and $\begin{bmatrix} -6 \\ -7 \end{bmatrix}$ were very common wrong answers.

Q10.

In part (a), most candidates were able to complete the square for at least one of the terms. A very common incorrect answer involved $(x - 7)^2$ instead of $(x + 7)^2$. Occasionally a negative answer was given for the square of the radius. Also, $\sqrt{5}$, $\sqrt{25}$ and ± 49 were often seen on the right-hand side of the equation.

In part (b), the coordinates of the centre usually followed through correctly from part (a). Often the correct radius was stated, even though part (a) was incorrect.

In part (c), the centre of the circle usually appeared in the correct quadrant for the candidate's coordinates. However, those who had the correct answers in part (b) did not always draw the correct circle touching the x -axis at -7 and not touching the y -axis.

In part (d)(i), several candidates earned the method mark but the most common error lay in the failure to square (kx) as k^2x^2 ; most wrote kx^2 and immediately lost the accuracy mark.

In part (d)(ii), candidates seemed to think it necessary to write " $= 0$ " only on the very last line. This does not fulfil the requirements of a proof. The first line should be that equal roots occur when $b^2 - 4ac = 0$ and the " $= 0$ " should continue throughout.

Other common errors occurred in the squaring of $2(k + 7)$, which often became $2(k + 7)^2$. Candidates should check each line of their proof instead of fudging one or more lines of working, as examiners are unlikely to be deceived by the miraculous appearance of the printed answer following incorrect working.

Many did not attempt part (d)(iii). The factorisation of the quadratic expression defeated most candidates; too many are reliant on the quadratic equation formula which often produces large numbers that are difficult to handle without a calculator. The incorrect factors $(4k - 3)(3k + 4)$ were seen as often as the correct ones. Candidates found this factorisation extremely hard and very few picked up these two final marks, but this is a topic that should not be beyond students at this level.

Q11.

In part (a), some weaker students did not seem familiar with the standard equation of a circle, with several writing the left hand side of the equation as $(x + 5)^2 + (y + 8)^2$ or $(5 - a)^2 + (8 - b)^2$. However, most errors were associated with the value of k ; typical incorrect values seen on the right hand side of the equation were 5, 89 and $\sqrt{89}$.

In part (b)(i), the verification mark was rarely scored. Many, who had the correct circle equation, were able to verify that A was a point on the circle but then neglected to make a concluding statement. Those who had the wrong value of k in part (a) could not earn this mark unless they argued in terms of Pythagoras, stating the correct value of the radius.

In part (b)(ii), while most students found the gradient of AC correctly, many used the value of the gradient of AC when finding the equation of the tangent. It was pleasing to see a large number of students realise the need to find the negative reciprocal, but these students were sometimes unable to present their tangent equation in the required form.

In part (c)(i), most used the distance formula correctly and found that CM^2 was equal to 20 and then deduced that $CM = 2\sqrt{5}$. Occasionally the surd was not handled correctly, with some writing the final answer as $4\sqrt{5}$. No marks were awarded to students who simply wrote down the answer $2\sqrt{5}$ without supporting working.

Part (c)(ii) was not answered quite so well. Pythagoras was not always used correctly as the angle PCQ at the centre of the circle was often assumed to be a right angle or the length CM assumed to equal PM . Even those who found the correct value of PM either found the area of only half the triangle or were unable to deal with the product of the surds.

Q12.

In part (a)(i) the LHS of this equation was almost always correct but the RHS was less commonly so with $\sqrt{13}$ or 13^2 being common wrong answers.

In (a)(ii) again, the first 4 terms, namely $x^2 + y^2 + 6x - 4y$, were generally correct with just the odd slip occurring, either in a sign or perhaps from not doubling one of the x or y coefficients. The rest of the equation was only completed correctly by about half of the candidates, although some managed to recover from an error in their equation in part

(a)(i), usually replacing $\sqrt{13}$ by 13.

Part (b) defeated many candidates. Using algebra, putting $x = 0$ and solving the quadratic to find the y -intercepts, and hence finding the length of AB , was done only by the most competent. However some candidates were successful using a diagram and the theorem

of Pythagoras. Unfortunately some who drew a picture assumed that the two radii to the intercepts were at right angles and their Pythagoras calculations were incorrect.

Part (c)(i) proved an insurmountable hurdle for those who tried to use their wrong circle equation; fortunately those who used the form given in the question were more likely to be successful. Some compared the distance from the point to the centre of the circle with the radius, but, in many cases, there was some confusion between 13 and $\sqrt{13}$ and these candidates lost the mark. Unfortunately some also lost the mark by failing to complete their verification with a relevant statement.

Part (c)(ii) was the only place in this question paper where it was essential to know how to find the gradient of the line joining two points. A few candidates had their expression upside down; however many who wrote down a correct expression for the gradient were often unable to cope with the negative signs when simplifying their gradient into a simple fraction.

In (c)(iii) the tan gent to the circle required the negative reciprocal of the gradient of CD to be calculated, since a perpendicular line was involved, but many seemed unaware of this procedure.

Q13.

In part (a) most candidates found the correct values of a and b , but correct values for k were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side.

In (b) some used an approach based on Pythagoras, but the majority of candidates used an algebraic method, substituting $y = 0$ into their circle equation and solving the resulting quadratic equation in x . Although there were a number who made sign errors, many candidates produced the correct values of x . Nevertheless, common mistakes such as putting $x = 0$ or $(y + 8) = 0$, instead of $y = 0$, resulted in no marks being earned.

In part (c) many candidates assumed that the gradient of CA was 2.5 instead of finding the negative reciprocal and many who did have a correct equation were unable to produce an equation with integer coefficients as requested. A sketch might have helped some to have seen the correct configuration and perpendicularity.

In (d)(i) most candidates realised the need to substitute $y = 2x + 1$ into one or other form of their circle equation but algebraic errors abounded as they tried to produce the displayed quadratic equation. Those who expanded the circle equation first were generally more successful; others who did not have 100 on the right hand side of their circle equation failed to recognise this error when they failed to produce the printed answer for the equation.

Candidates should be strongly discouraged from starting each new line with an equals sign when producing an algebraic proof of this type that involves equations.

In (d)(ii) those who completed the square failed to score full marks if they simply wrote $x + 3 = \sqrt{11}$ before concluding that $x = -3 \pm \sqrt{11}$. Those using the quadratic equation formula

often struggled to simplify $\frac{-6 \pm \sqrt{44}}{2}$.

Q14.

In part (a), most candidates found at least one of the correct coordinates for the centre C , with the most common error being at least one sign error or writing the centre as $(4, -12)$.

However, the correct value for the radius was not so common, with $\sqrt{15}$ being frequently seen.

In part (b), most explanations involving a comparison of the y -coordinate of the centre and the radius of the circle earned at least one mark. In order to score the second mark, some of the better answers explained why the y -coordinate of the “highest” point of the circle was -1 , but most comments were insufficient. Those who proved that the circle did not intersect the x -axis needed to state that the y -coordinate of the centre was negative in order to score full marks.

Part (c)(i) was very poorly done. Many candidates tried to use their circle equation and essentially faked the given result instead of correctly using the distance formula. Once again, if candidates are asked to “show that ...” then the full equation needs to be seen in the final line of the proof and an essential part of the working was to see a statement such as $PC^2 = 3^2 + (k + 6)^2$, leading to the printed answer.

In part (c)(ii), the word “hence” indicated that the expression for PC^2 in part (c)(i) needed to be used. Candidates needed to realise that the point P lies outside the circle when $PC > r$ and that using this result immediately leads to the given inequality. The inequality sign here caused confusion with many looking for a discriminant.

In part (c)(iii), many candidates scored only the two marks available for finding the critical values -2 and -10 . No doubt some would have benefited from practising the solution of inequalities of this type by drawing a suitable sketch, or by familiarising themselves with the technique of using a sign diagram as indicated in previous mark schemes.

Q15.

Part (a) Most candidates obtained the correct left hand side of the circle equation but many failed to recognize that the radius was 5. Alarming many thought that r was equal to -5 or wrote down the right hand side of the equation as -5^2 , thus displaying a fundamental misunderstanding of the idea of radius as a length.

Part (b)(i) Most who had the correct circle equation were able to verify that the circle passed through the point P , although those who neglected to make a statement as a conclusion to their calculation failed to earn this mark.

Part (b)(ii) The negative signs caused problems for many when finding the gradient of PC and only the better candidates obtained the correct value. Many candidates then found the negative reciprocal of this fraction instead of using the gradient of PC to find the normal to the circle at the point P .

Part (b)(iii) There were basically two approaches to this question, although some candidates were merely guessing and no credit was given for a correct answer without

supporting working.

The most common method involved distances or squares of distances; many made errors in finding the coordinates of M and then struggled with the fractions when squaring and adding to find the length of PM ; whereas others noted that the length of PM was simply half the radius. A simple comparison with the length of PO led to the correct conclusion.

The second approach was essentially one using vectors or the differences of coordinates, but this method was not always explained correctly and left examiners in some doubt as to whether candidates really understood what they were doing. The best candidates wrote down the correct vectors $\overrightarrow{PM}$ and $\overrightarrow{OP}$ and reasoned that these vectors had the same y -component but different x -components and it was then easy to deduce that P was closer to the point M .

Q16.

In part (a) most candidates found the correct values of a and b , but correct values for r^2 were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side. It was common to see things such as $25 = \sqrt{25} = 5^2$ and this could be penalised in the future.

In part (b) the coordinates of the centre, C and the radius r , although not always correct, usually gained full credit when following through from part (a).

In part (c)(i) most candidates attempted to verify that the point D was on the circle, although some, who had obviously worked a previous examination question, were keen to show that the distance from C to D was less than the radius and that D lay inside the circle. This verification was marked fairly strictly and the argument had to be correct including a final concluding statement. Those who simply wrote $4^2 + 3^2 = 25$, for example, did not earn the mark.

In part (c)(ii) many candidates found the gradient of CD and then assumed they had to find the negative reciprocal of this since the question asked for the normal at D . Reference to a sketch might have prevented this incorrect assumption.

In part (d)(i) most candidates made errors by not using brackets; the expression kx^2 was seen almost as often as the correct form k^2x^2 after substituting $y = kx$ into their circle equation.

In part (d)(ii), although there were some correct solutions seen, the discriminant often contained algebraic slips and the condition for equal roots was rarely stated. Often it was several lines into the working before an “= 0” appeared and many times this was omitted entirely. The value $k = 0$ was often ignored in otherwise correct solutions, but it was more common to see a three term quadratic because of previous algebraic errors.

In part (d)(iii) several candidates realised that the line would be a tangent for each of the two values of k , but many completely missed the point and talked about transformations, often giving vectors in their answer.

Q17.

In part (a)(i) most candidates realised what the correct coordinates of the centre were, although some wrote these as $(-5, 12)$ instead of $(5, -12)$.

Some gave the radius as 169 and others evaluated $\sqrt{169}$ incorrectly in part (a)(ii). The majority of candidates obtained the correct value of the radius.

In part (b)(i) most were able to verify that the circle passed through the origin, although some neglected to make a statement as a conclusion to their calculation and so failed to earn this mark. A surprisingly large number made no attempt at this part.

Most sketches were correct in part (b)(ii), though some were very untidy with some making several attempts at the circle so the diagram resembled the chaotic orbit of a planet. In spite of being asked to verify that the circle passed through the origin many sketches did not do so. Credit was given for freehand circles with the centre in the correct quadrant and which passed through the origin, although it was good to see some circles drawn using compasses. Many used algebraic methods, putting $x = 0$, but often their poor algebra prevented them from finding the value of p . Those using the symmetry, doubling the y -coordinate, were usually more successful, although an answer of -25 (from $-12-13$) was common.

In part (c)(i) the majority of candidates tried to find the gradient of AC but careless arithmetic meant that far fewer actually succeeded in finding its correct simplified value.

In part (c)(ii), in order to find the tangent, it was necessary to use the negative reciprocal of the answer from part (c)(i) in order to find the gradient. Although some did, many chose to use the same gradient obtained in the previous part of the question and scored no marks at all.

Q18.

In part (a), it was only necessary to complete the square for the y -terms. As a result, there were probably fewer errors this year expressing the left-hand side of the equation of the circle as $(y - 5)^2$. However, the right hand side was often written as $\sqrt{5}$, -5 or -45 instead of 5.

In part (b), quite a number who had the correct circle equation in part (a) wrote the coordinates of the centre as $(5, 0)$ or $(0, -5)$. Generous follow through marks were awarded for the radius provided the right-hand side of the equation had a positive value.

The wording in the question reassured most, though, that the radius was $\sqrt{5}$.

In part (c)(i), those with poor algebraic skills, often writing $2x^2$ instead of $(2x)^2$, struggled to establish the given quadratic equation. Also, quite a few made errors in their working but miraculously wrote down the given equation on their final line. A surprising number derived an equation in y . Quite a few simply solved the given quadratic equation in this part and thus failed to show an understanding of what was required.

In part (c)(ii), it was necessary to state that the equation had a repeated root of $x = 2$, or to use the zero value of the discriminant to show that the equation had equal roots, and hence to conclude that the line was a tangent to the circle.

In part (d), far too many simply substituted the coordinates of the point Q into the equation of the circle obtaining a nonsensical statement such as “ $-3 = 0$ so the point lies inside the circle”. It was necessary to see that the distance CQ was being calculated and then concluded that this distance was less than the radius of the circle, and hence the point Q must lie inside the circle.

Q19.

In part (a), while many found the centre and hence the values of a and b correctly, far fewer were able to find the radius. A very common incorrect answer was $\sqrt{233}$ instead of 13. A few candidates substituted the coordinates of the centre into the given equation so wrote $(8 - a)^2 + (13 - b)^2 = \dots$ and earned no marks. Too many candidates were drawn into fruitless calculations when all they needed to do was refer to the diagram.

In part (b)(i), the gradient of PC was usually correct.

The most common error in part (b)(ii) was a misunderstanding of what was required. Many assumed that the tangent was the straight line passing through P and C and used their answer to part (b), instead of its negative reciprocal, as the gradient. Not all of those who found the correct equation converted it to a form with integer coefficients.

In part (b)(iii), a surprising number of candidates did not realise that the distance from C to P was the radius, and recalculated it. It was a pity that they did not return to part (a) and correct their circle equation.

Those who drew a sketch usually used Pythagoras's Theorem correctly, but many failed to recognise that the distance required was the length of the perpendicular from the centre of the circle to the midpoint of the chord. A few obtained the correct answer by considering a horizontal chord and finding the point of intersection with the circle.

Q20.

Part (a) The $+ 2x$ term was ignored by many who wrote the left hand side of the circle equation as $(x - 1)^2 + (y - 6)$ but most candidates were able to complete the square correctly. The right hand side was often seen as 49 and since this was a perfect square it did not cause candidates to doubt their poor arithmetic.

Part (b) Many who had the correct circle equation in part (a) wrote the coordinates of the centre with incorrect signs. Generous follow through marks were awarded in this part provided the right hand side of the equation had a positive value.

Part (c) Almost all candidates reasoned correctly by considering the y -coordinate of the centre and the radius of the circle, although a number were successful in showing that the quadratic resulting from substituting $y = 0$ into the circle equation does not have real roots. Some simply drew a diagram and this alone was not regarded as sufficient to prove that the circle did not intersect the x -axis. Others using an algebraic approach found a quadratic that they said did not factorise and concluded incorrectly that the equation had no real roots.

Part (d) The algebra proved too difficult for the weaker candidates and many who had shown good algebraic skills rather casually forgot to include “ $= 0$ ” on their final line of

working. Sadly, many were unable to factorise the quadratic or wrote the coordinates of Q as $(-5, 2)$. It was good; however, to see more candidates being able to find the correct mid-point, where in previous years too many had found the difference of the coordinates before dividing by 2.

Q21.

Part (a) Most candidates found the correct coordinates of the centre, although some wrote these as $(3, -2)$ instead of $(-3, 2)$. Those who multiplied out the brackets were often unsuccessful in writing down the correct radius of the circle.

Part (b)(i) Most candidates were able to verify that the point N was on the circle, although some, who had perhaps worked a previous examination question, were keen to show that the distance from C to N was less than the radius and that N lay inside the circle.

Part (b)(ii) Most sketches were correct, though some were very untidy with several attempts at the circle so that the diagram resembled a chaotic orbit of a planet. Some candidates omitted the axes and scored no marks.

Part (b)(iii) The majority of candidates found the gradient of CN and then assumed they had to find the negative reciprocal of this since the question asked for the normal at N . Reference to their diagram might have avoided this incorrect assumption.

Part (c)(i) Most wrote $PC^2 = 5^2 + 4^2$, provided they had the correct coordinates of C .

However, the length of PC was often calculated incorrectly with answers such as $\sqrt{31}$ and $\sqrt{36} = 6$ seen quite often.

Part (c)(ii) Although there were many correct solutions seen, Pythagoras' Theorem was often used incorrectly. A large number of candidates wrote the answer as a difference of two lengths such as $\sqrt{41} - 5$. Candidates need to realise that obtaining the correct answer from incorrect working is not rewarded; quite a few wrote $\sqrt{41} - \sqrt{25} = \sqrt{16} = 4$ and scored no marks. Many who drew a good diagram realised that a tangent from $(2, 6)$ touched the circle at $(2, 2)$ and so the vertical line segment was of length 4 units.

Q22.

- (a) Completion of the squares in the circle equation was carried out well once more. The most common error was a sign slip usually in the second term. Another error lay in combining the constant terms, so answers such as -14 and 11 were seen for r^2 .
- (b) Most earned the mark for the coordinates of the centre of the circle as this was a follow through mark. The mark for the radius was not always earned as some failed to take the square root or had an inappropriate answer such as a negative value for r^2 .
- (c)
 - (i) Most found CO to be 5. However, a few neglected to square -3 and 4 before adding and some subtracted 9 from 16.
 - (ii) This part was answered well with most realising the need to explain, using both lengths, why O lay inside or outside the circle. Some accompanied their

explanations with diagrams, although this was not necessary.

Q23.

Part (a)(i) It was apparent that some candidates had not covered this part of the specification and they made no progress. Most who had done so, earned the marks here. However a few wrote $(x + 2)^2$ or even $(x - 2)^2$ and then gave the centre as $(-2, 0)$. Some managed to incorporate the 7 with the y term so wrote the coordinates of the centre as $(2, -7)$.

Part (a)(ii) Most candidates were successful in finding the correct radius. However some forfeited one mark by 'meddling' with their equation and putting 18^2 or $\sqrt{18}$ on the right hand side of the equation.

Part(b) This part was rarely attempted. Even where a correct diagram was drawn, few recognised that the chord would be bisected. Many assumed that the triangle was right-angled at the centre of the circle. Others drew tangents instead of a chord.

Part (c)(i) Many made little progress here. However, it was good to see more able candidates coping well. A few fell at the final line writing $(k - 1)$ instead of $(k + 1)$; a few lost a mark by not introducing ' $= 0$ ' as part of the equation of the circle and simply added ' $= 0$ ' at the end of several lines of working so as to match the printed answer. Many made a slip in squaring $(2k - x)$ and some made gross errors such as writing this as $4k^2 + x^2$ or $4k^2 - x^2$. Others 'simplified' the equation to $(x - 2) + (2k - x) = \sqrt{18}$.

Part (c)(ii) Those candidates who made progress here needed both knowledge and algebraic skills and only a small minority completed this part correctly. However more earned some method marks. Use of the correct condition on the discriminant was required but some just tried to solve the equation using the quadratic formula or used ' > 0 ' instead of ' $= 0$ '. A few attempts at completing the square were seen but most failed to equate the expression to zero.

Part (c)(iii) Many candidates who had made no progress in the rest of the question stated that the line would be a tangent to the circle; however several candidates wrote at length about various transformations and completely missed the point.

Q24.

- (a) Most candidates seemed aware of the need to complete the square, but sign errors were very common when finding r^2 .
- (b) Provided r^2 was given as a positive value, marks were awarded for following through from their answers to part (a).
- (c) Many did not have the stamina to obtain the printed quadratic equation giving the x -coordinates of P and Q . They could usually solve the quadratic equation, but it was very common to see the coordinates of P and Q given as $(2, 0)$ and $(3, 0)$ instead of substituting into the equation $y = x + 4$ to find the correct values of y .

Q25.

Part (a) The equation of a circle seemed widely known, although some candidates wrote

things such as $C = 2\pi r$. Sign errors were common and some forgot to square the radius or wrote $\sqrt{5}$ on the right side of the equation.

Part (b) Most candidates realised that the easiest way to verify that the point P lies on the circle was to substitute $x = 6$ and $y = 2$ into the left hand side of the equation and to show this was equal to 25, the same as the right hand side. Other ingenious methods were used in addition to verifying that PC was equal to 5.

Part (c) The same errors relating to part (c)(i) of question 1 were repeated here. Most candidates, however, were successful in finding the correct gradient of CP.

Part (d)(i) The relationship that the product of the gradients is -1 seems well known but a number of arithmetic slips occurred when finding the gradient of the perpendicular line.

Part (d)(ii) Several ignored the hint of part (d)(i) and many failed to realise that the tangent was a line passing through P(6, 2). It was not necessary to simplify a correct form of the equation in order to score full marks. Those who used $y = mx + c$ usually fared badly here, because they rarely obtained a correct form of the equation.



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